QUT Centre for Robotics



ENN583 – Foundations of Robotic Vision

Week 13 – Sem 2 – 2026

Prof Niko Suenderhauf

Welcome to Week 13!



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From Computer Vision to Robotic Vision

Reliability of Deep Learning in Robotics
What is the Problem?

The cost of mistakes can range from annoying to life-threatening.





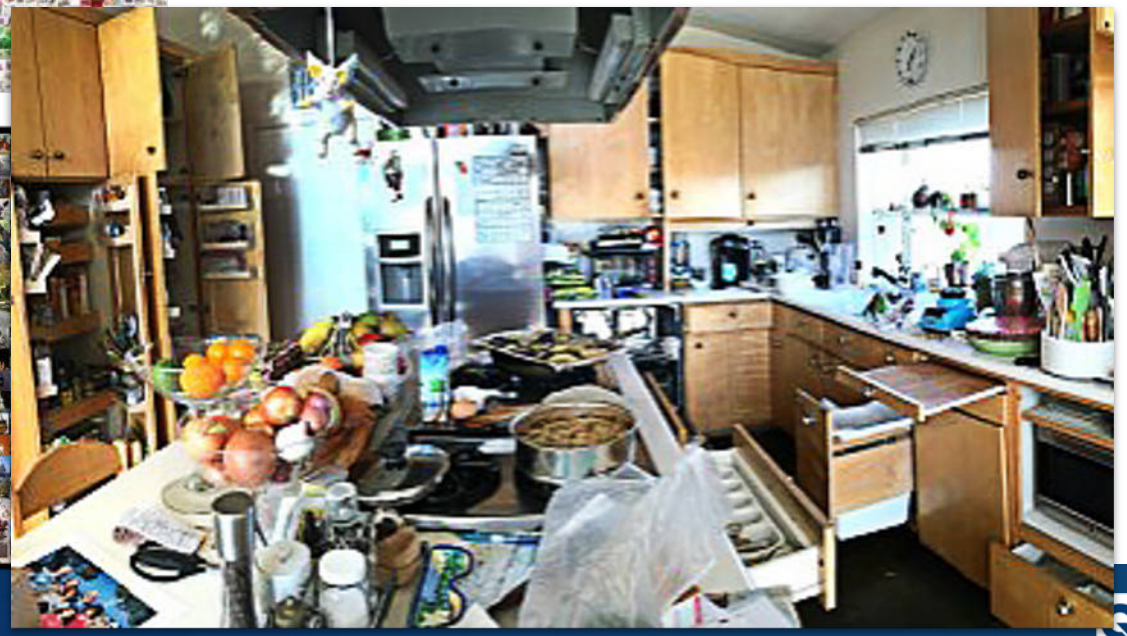
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Reliability of Deep Learning in Robotics
What is the Problem?

Training under **Closed-Set** conditions.
Deployment under **Open-Set / Out-of-Distribution** conditions.



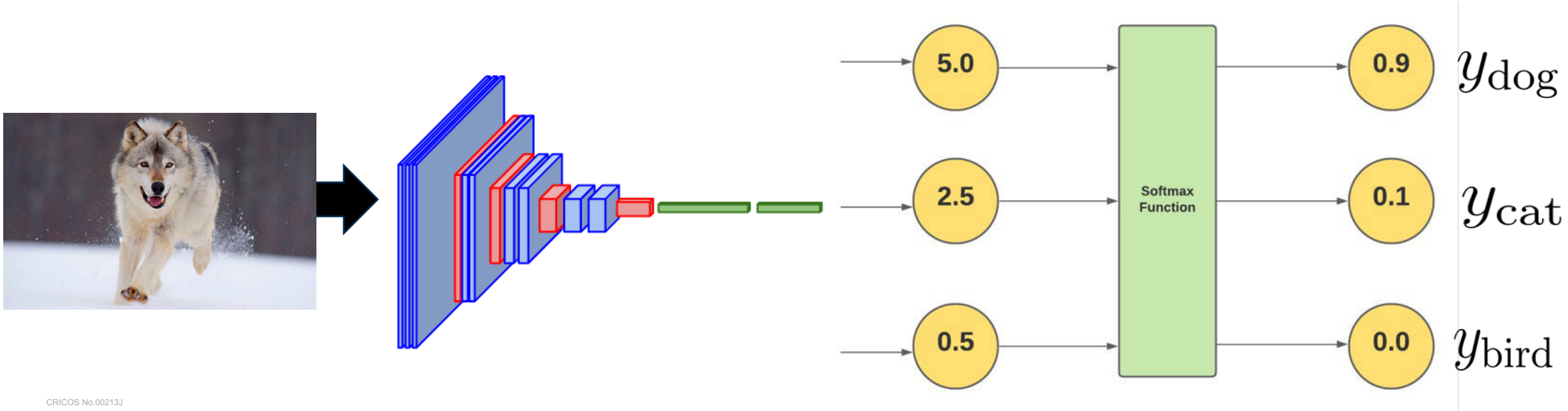




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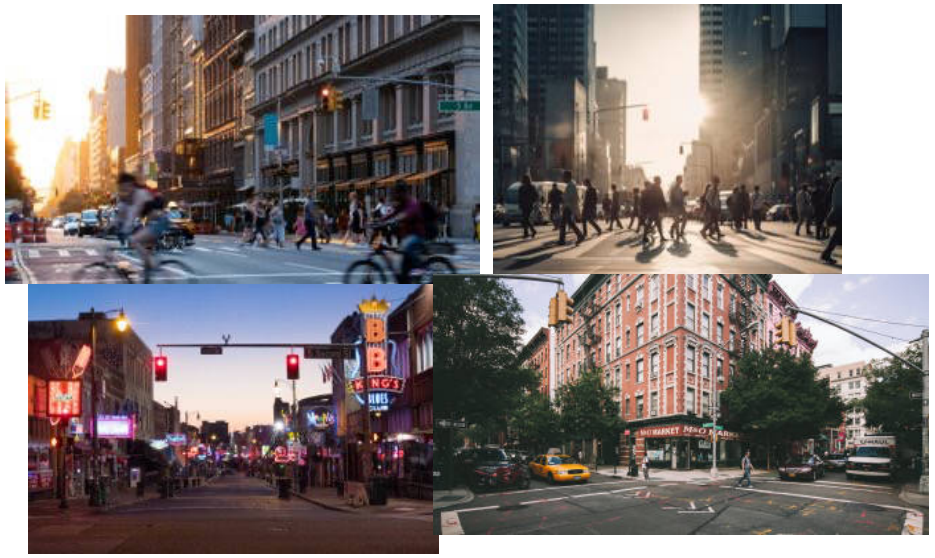
Reliability of Deep Learning in Robotics
What is the Problem?

No clear mechanism for handling new classes (classes not seen during training).

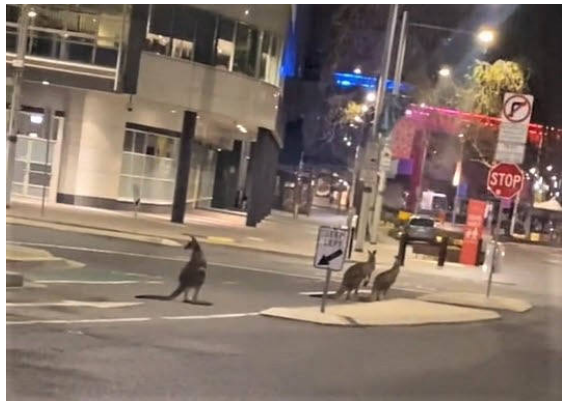


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Reliability of Deep Learning in Robotics
What is the Problem?



Training data



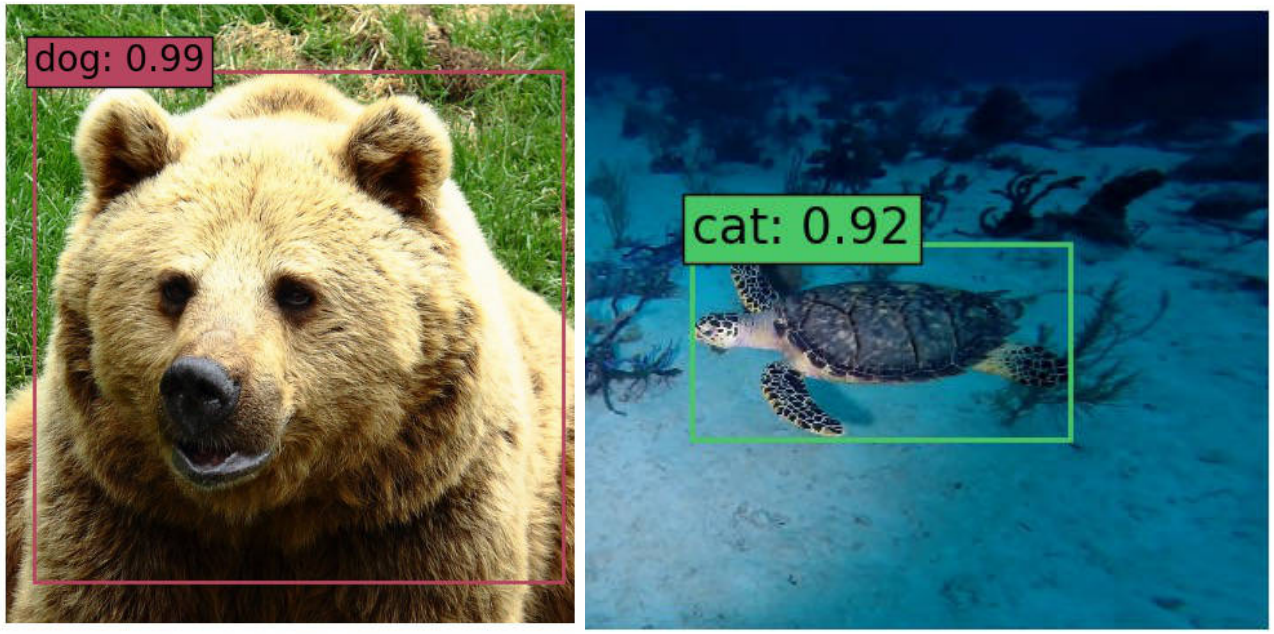


New objects appear during deployment.
“Open Set”

Reliability of Deep Learning in Robotics
What is the Problem?

Typical failure: unknown objects are confidently mis-classified as one of the known classes.

Image credit: Dimity Miller



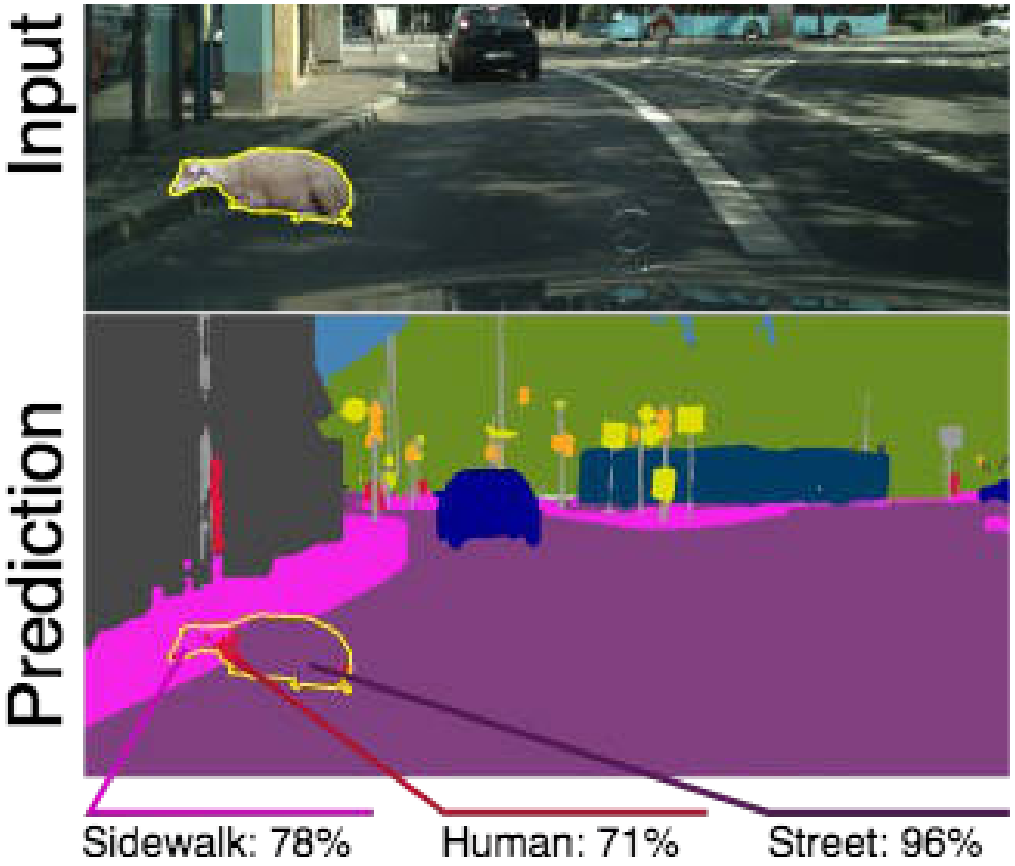
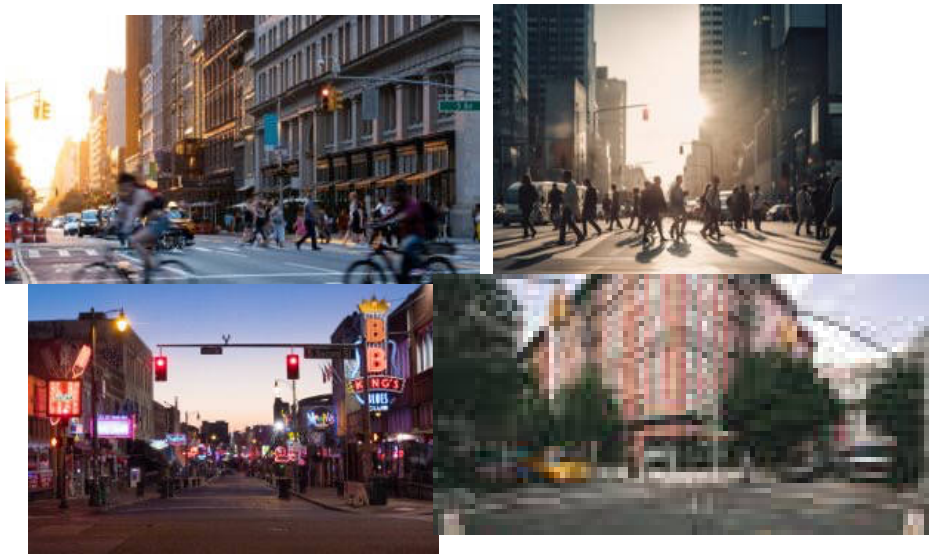
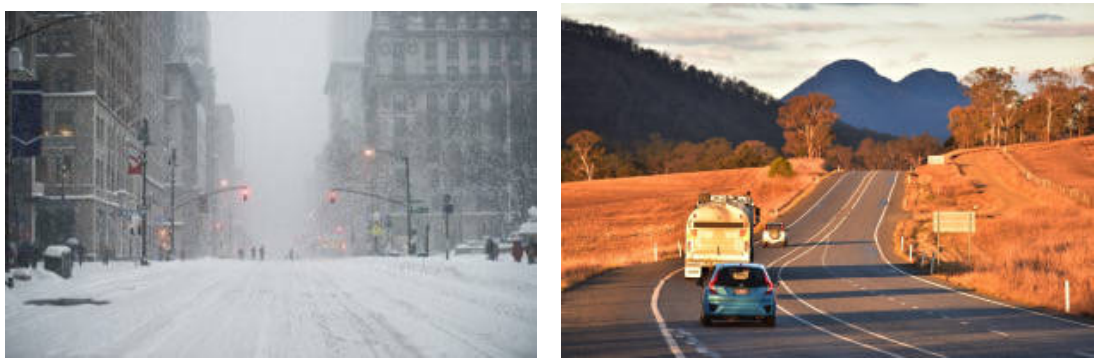


Image credit: Hermann Blum et al. https://fishyscapes.com/

Reliability of Deep Learning in Robotics
What is the Problem?



Training data



Changes to conditions → changes to features.
"Out of distribution"

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Different Problems in the Literature

Out-of-distribution (OOD):

- An input comes from a different distribution than the data used to train the model.
- Formally, deployment includes x from regions where $p_{\text{train}}(x)$ is very small or zero.

Open-set recognition:

- At test time, the model encounters samples from classes not seen during training.
- It should recognise them as unknown rather than force them into a known class.

Open-world recognition:

- extends open-set recognition by allowing the system to encounter unknown classes, detect them as unknown, and then learn/incorporate new classes over time.

Beware: these terms overlap but are not identical. An OOD sample does not have to belong to a novel semantic class; it could simply be a known class under very different conditions, such as heavy fog or a new sensor. Conversely, open-set problems specifically focus on unknown classes.

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Core Assumption in Machine Learning

Training and deployment data are assumed to come from the **same joint distribution** $p(x, y)$**.**

$$p_{\text{train}}(x, y) = p_{\text{deploy}}(x, y)$$

(x = data and y = labels)

Distribution shift occurs when this assumption is violated.

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Types of Distribution Shift

(x = data and y = labels)

- **Covariate Shift:** $p_{\text{train}}(x) \neq p_{\text{deploy}}(x)$ $p(y \mid x)$ stays the same
  - e.g. change in lighting, weather, camera, environment, corruption
- **Label Shift** / Prior Shift: $p_{\text{train}}(\text{y}) \neq p_{\text{deploy}}(\text{y})$ $p(x \mid y)$ stays the same
  - e.g. some classes become much more/less common
- **Concept Shift:** $p_{\text{train}}(y \mid x) \neq p_{\text{deploy}}(y \mid x)$
  - the mapping from appearance to label changes
- **Novel classes** / Open-Set

$$\exists (x, y^*): p_{\text{train}}(x, y^*) = 0 \quad \text{but} \quad p_{\text{deploy}}(x, y^*) > 0$$

  - Inputs belong to classes never seen during training

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Reliability of Deep Learning in Robotics
What is the Problem?

Models often output a **softmax distribution.**
These values look **like probabilities.**
But they are **not guaranteed to match the model's true likelihood of being correct.**

Why this matters

- Downstream systems may use confidence to:
  - fuse predictions from multiple sensors or models
  - decide whether to trust a perception result
  - trigger fallback or human intervention
  - weight measurements in a larger estimation pipeline
- If confidence is miscalibrated, downstream decisions can be systematically wrong.

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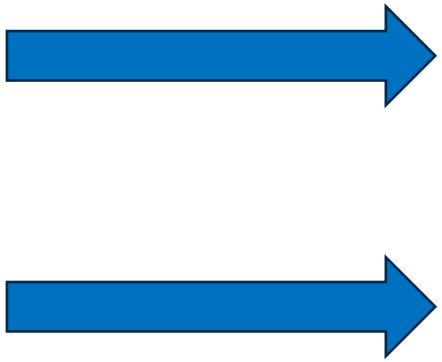
When should a robot trust a prediction?

Uncertainty Estimation

- "How unsure is the model about its prediction?"
- Model-centric signal

OOD Detection

- "How unfamiliar is the input?"
- Data-centric signal



Failure / Risk estimation

- "Should the robot trust this prediction?"
- Decision-centric question

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Let’s take a look at solutions:

- Confidence Calibration
- Uncertainty Estimation
- Out-of-Distribution detection

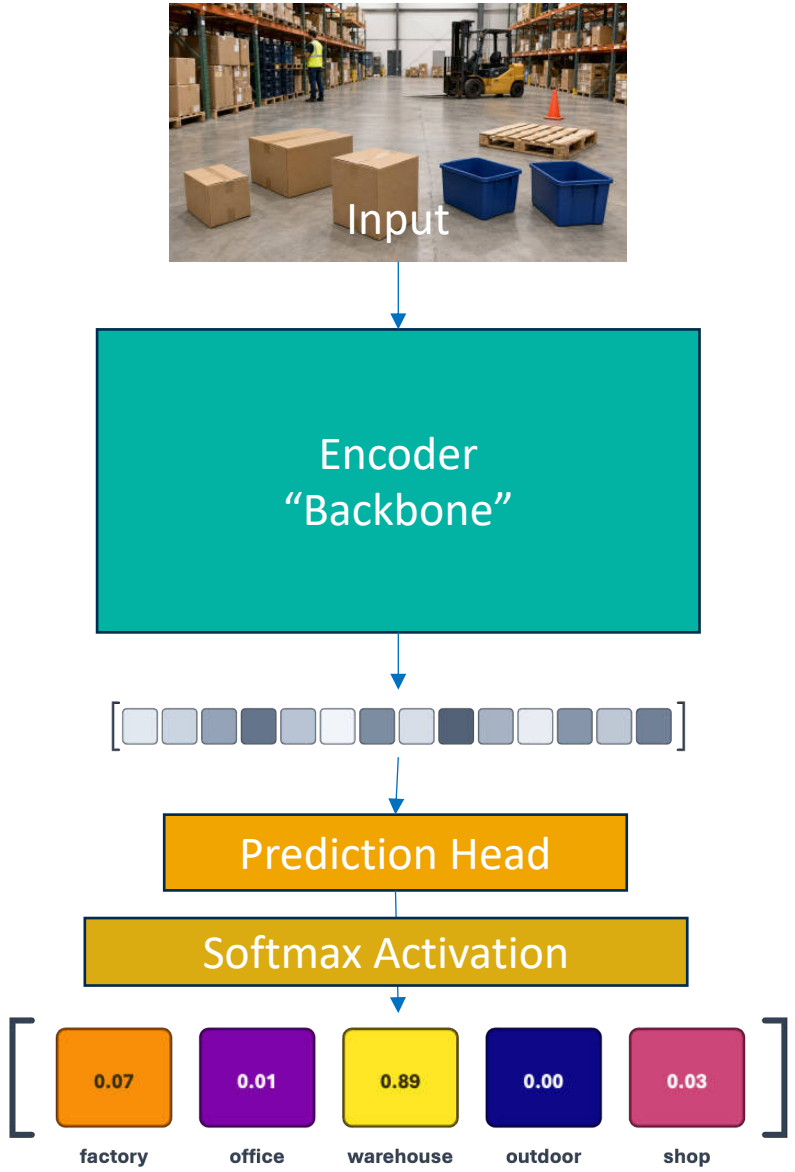
- Decision-making

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Is ”Confidence” Enough?

Prediction Confidence

- Models with softmax activations output “confidences” that look like a probability distribution
  - i.e. values between 0 and 1, and sums to 1
- But how meaningful are they?
  - Are they correlated with correctness?
  - What should the ideal behaviour be?



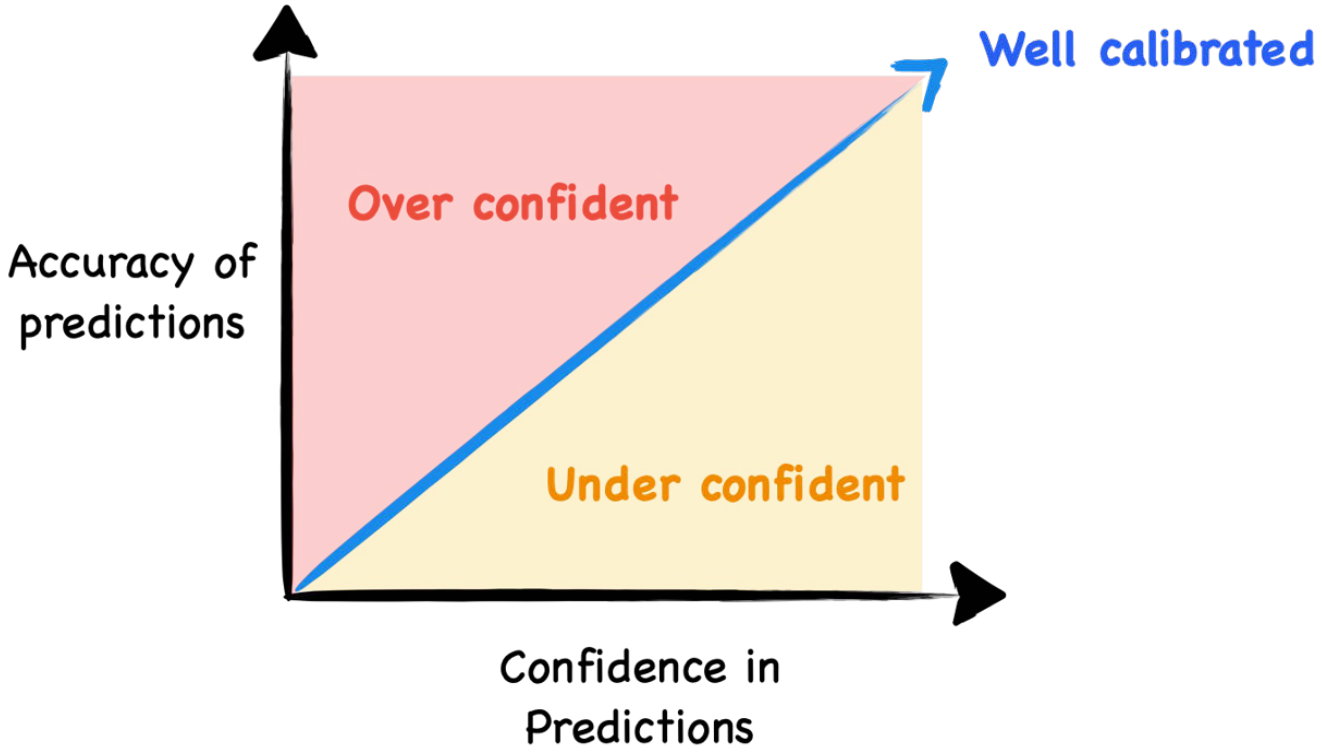
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What Should 80% Confidence Mean?

For an ideal (“calibrated”) classifier:

- **n% of predictions with n% confidence should be correct**

- E.g. 80% of the images classified with 80% confidence as “dog” should actually be “dog”



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Measuring Confidence Calibration

1. **Test the model** on the dataset, and for every prediction store:
    1. If it was correct or incorrect
    2. Model confidence

2. **Divide all the possible confidence values into N bins**
   (e.g. 0-10%, 10-20%, 20-30%, …., 80-90%, 90-100%).

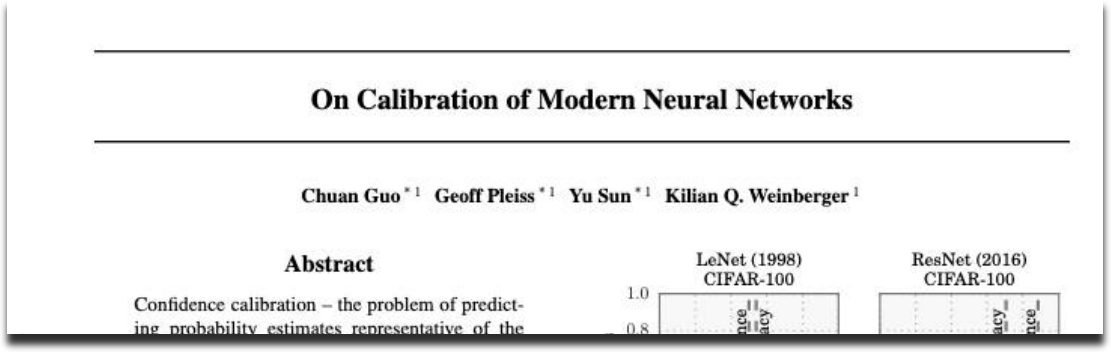
3. **For each bin:**
    1. Find the predictions in that confidence range (e.g. 0-10%)
    2. Calculate the mean accuracy of all those predictions
    3. Record on the plot



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How well calibrated are today's models?

Modern neural networks can be highly accurate – but poorly calibrated.

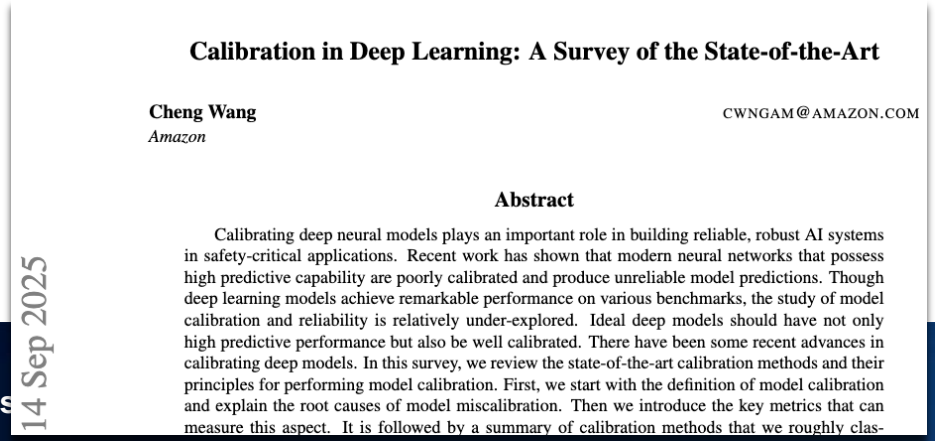


On Calibration of Modern Neural Networks

Chuan Guo [*1] Geoff Pleiss [*1] Yu Sun [*1] Kilian Q. Weinberger [1]

Abstract

Confidence calibration – the problem of predict-



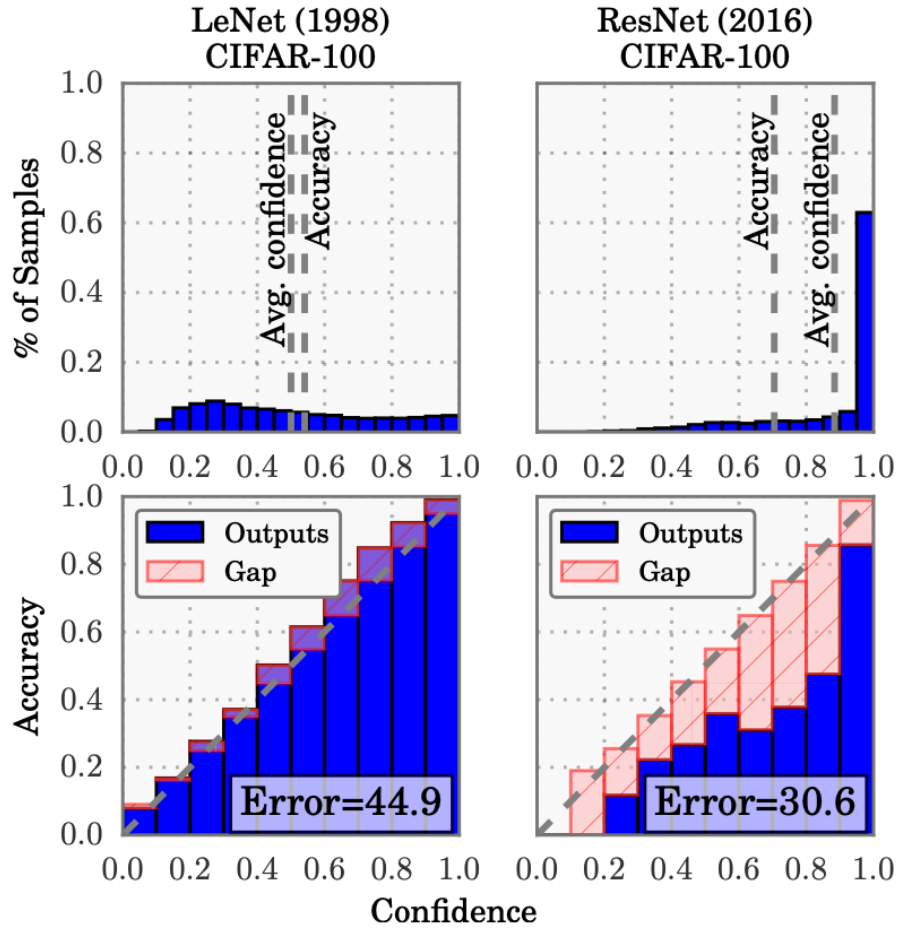
Calibration in Deep Learning: A Survey of the State-of-the-Art

Cheng Wang
Amazon
CWNGAM@AMAZON.COM

Abstract

Calibrating deep neural models plays an important role in building reliable, robust AI systems in safety-critical applications. Recent work has shown that modern neural networks that possess high predictive capability are poorly calibrated and produce unreliable model predictions. Though deep learning models achieve remarkable performance on various benchmarks, the study of model calibration and reliability is relatively under-explored. Ideal deep models should have not only high predictive performance but also be well calibrated. There have been some recent advances in calibrating deep models. In this survey, we review the state-of-the-art calibration methods and their principles for performing model calibration. First, we start with the definition of model calibration and explain the root causes of model miscalibration. Then we introduce the key metrics that can measure this aspect. It is followed by a summary of calibration methods that we roughly clas-

14 Sep 2025



How do we calibrate a neural network?

Temperature scaling

- Given logits z_i, standard **softmax** is

$$p_i = \frac{e^{z_i}}{\sum_j e^{z_j}}$$

- Temperature scaling introduces one scalar T

$$p_i = \frac{e^{z_i/T}}{\sum_j e^{z_j/T}}$$

- T>1: softens predictions, useful for an overconfident model
- T<1: sharpens predictions, useful for an underconfident model

- Learn T on a held-out validation set
  - **Important:** calibration learned on in-distribution validation data may not remain valid under **distribution shift**.

- Because all logits are scaled equally, the predicted class does not change
- Very cheap and simple to apply

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What tells us that a prediction may be unreliable?

- **Confidence** is the simplest signal – but not enough
  - maximum softmax probability / detector score (often surprisingly good)
  - → reject predictions with **low maximum confidence** or **high entropy**

Neural networks can still be **confidently wrong**,
especially under distribution shift!

Confidence is useful only if it is well calibrated.
But even then it is not a reliable OOD detector.

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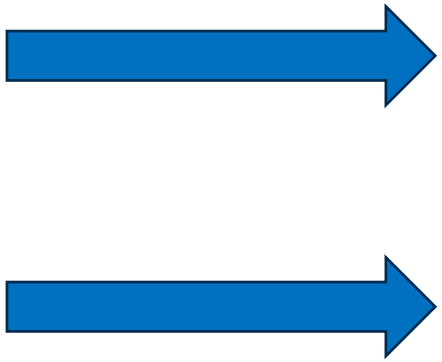
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Failure / Risk estimation

- "Should the robot trust this prediction?"
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Uncertainty

What is this? – An Experiment

What is this?



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What is this?



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What is this?



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What is this?



Two Types of Uncertainty



Epistemic Uncertainty

- Due to lack of knowledge
- Can be reduced by more data.

Aleatoric Uncertainty

- Due to noise inherent in the observations
- Can **not** be reduced with more data.



What Uncertainties Do We Need in Bayesian Deep Learning for Computer Vision? Alex Kendall and Yarin Gal, NeurIPS 2017.

Aleatoric Uncertainty

- Train the network to predict an uncertainty term σ_i^2 for each prediction $\widehat{x_i}$
- Use a loss that down-weighs the prediction error according to σ_i^2, but also penalises high σ_i^2

$$\mathcal{L} = \frac{1}{D}\sum_i \left[\frac{|x_i - \widehat{x_i}|^2}{2\sigma_i^2} + \frac{1}{2}\log \sigma_i^2\right]$$

- The network learns both the prediction and how much that prediction should be trusted.

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What Uncertainties Do We Need in Bayesian Deep Learning for Computer Vision?
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How can Uncertainty be Estimated?

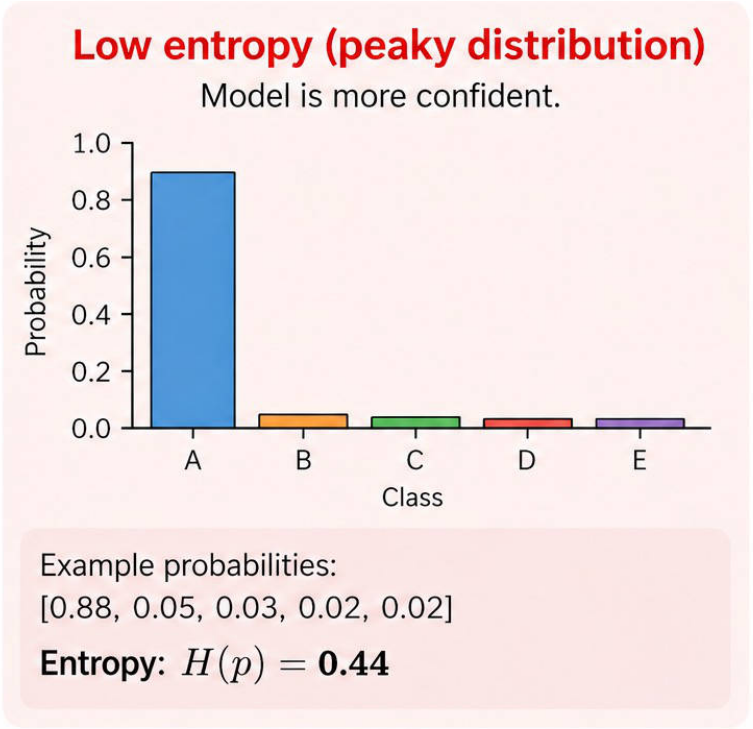
- looks at the model's confidence/disagreement for a specific prediction
- can be high even for an in-distribution but ambiguous image
- techniques:
  - predictive entropy
  - ensemble disagreement
  - Monte Carlo dropout

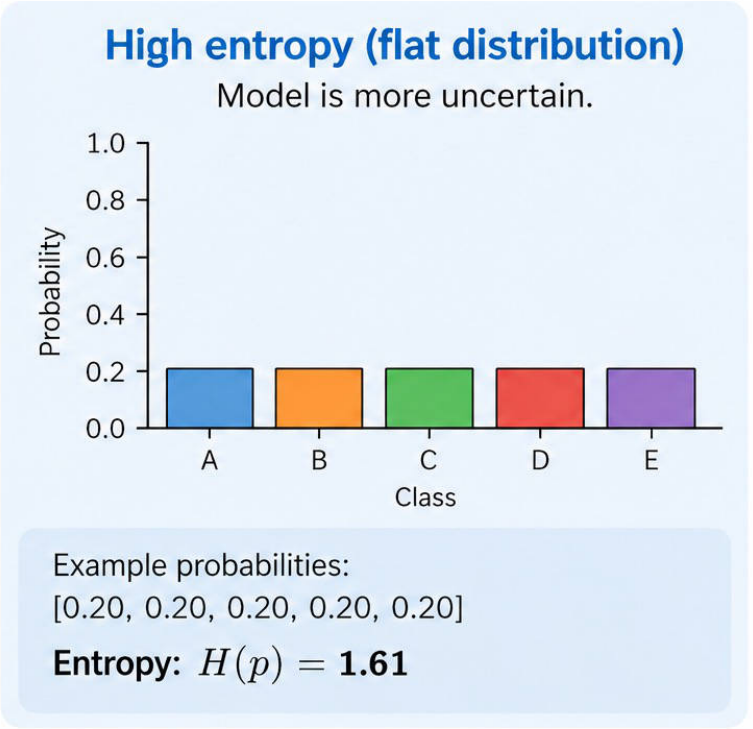
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How can Uncertainty be Estimated?

- Most simple: predictive **entropy:** How “peaky” is the distribution?
  - High entropy → probability spread across classes → more uncertainty
  - Low entropy → one class dominates → less uncertainty

$$H(\mathbf{p}) = -\sum_{k=1}^{K} p_k \log p_k$$



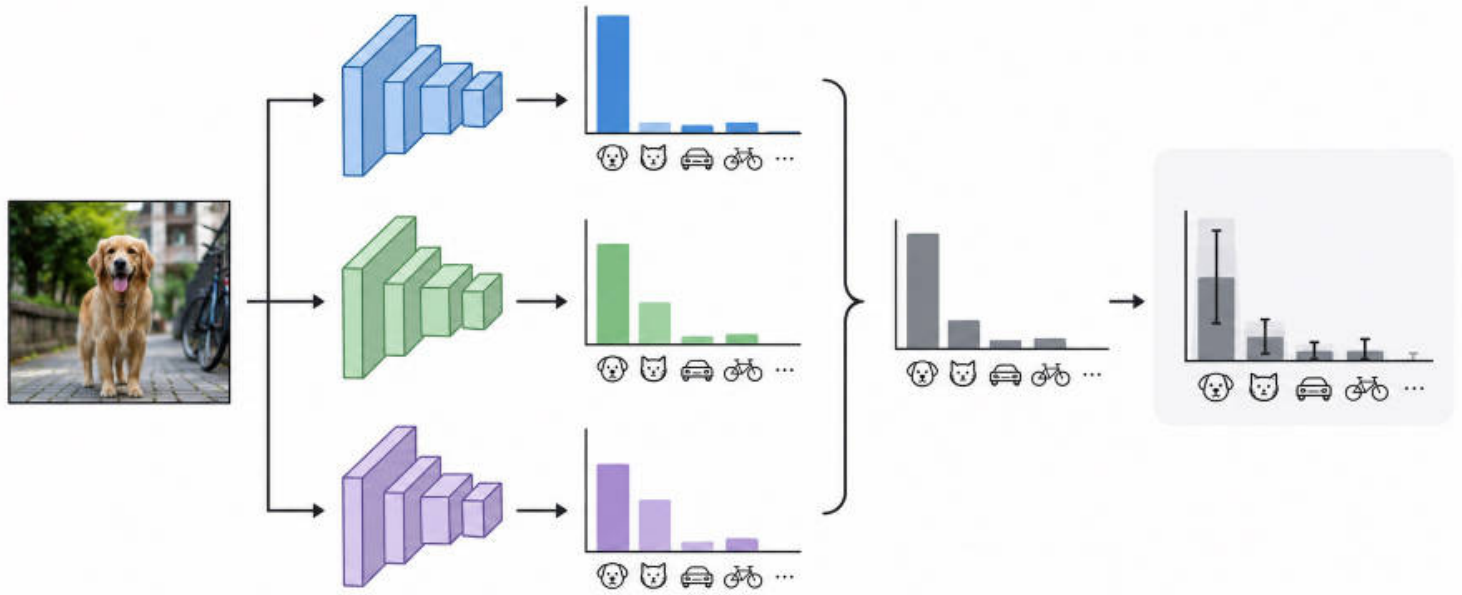


How can Uncertainty be Estimated?

Deep Ensembles

- Key idea: If different plausible versions of the model disagree. the model should be less certain.

- Train the same model multiple times, starting from different initialisations
- At test time: **average predictions across models**
- **Disagreement between models** provides an estimate of uncertainty
- Particularly useful for capturing **epistemic uncertainty**

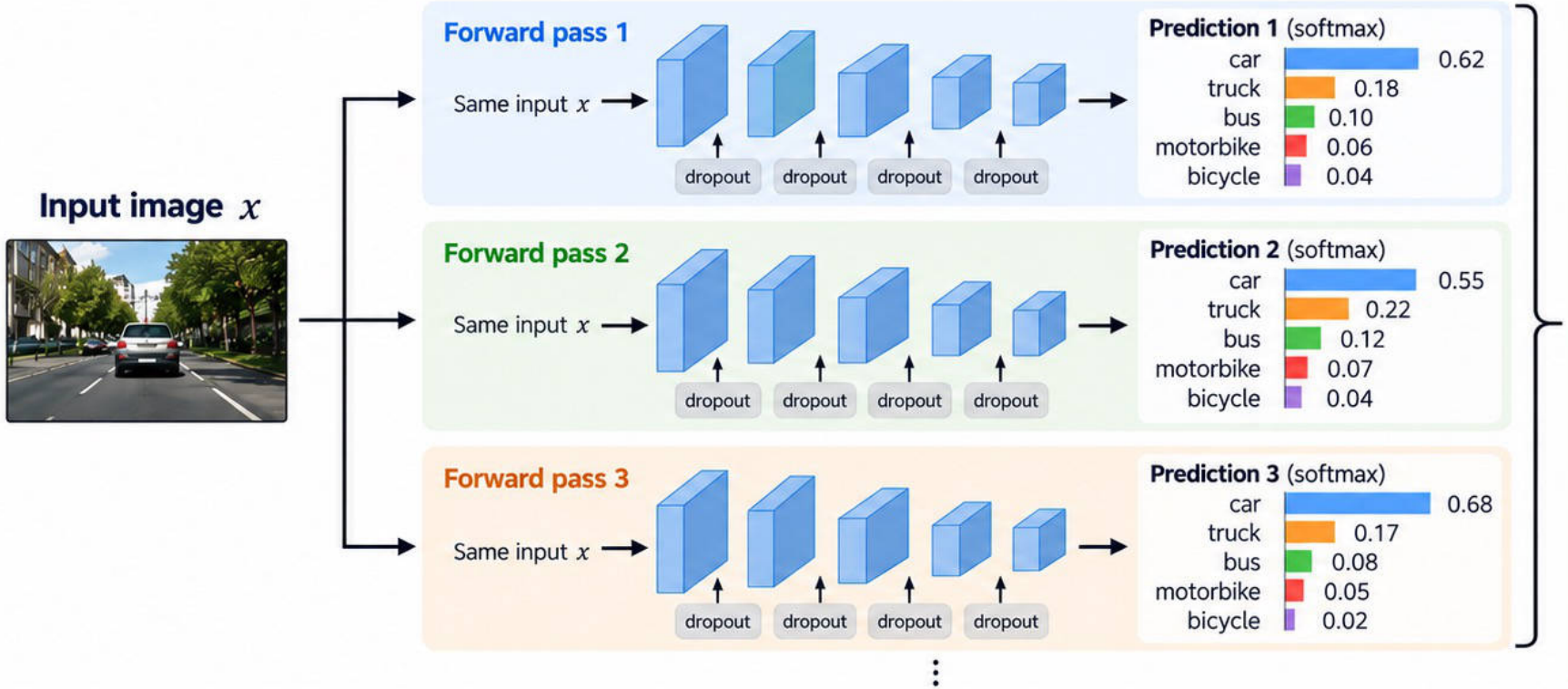


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How can Uncertainty be Estimated?

Monte Carlo Dropout

- Key idea: If different plausible versions of the model disagree, the model should be less certain.
  - Use dropout at test time and run the **same input multiple times.**
  - Each forward pass uses a **different random dropout mask.**
  - The resulting variation in predictions reflects **model uncertainty (epistemic).**

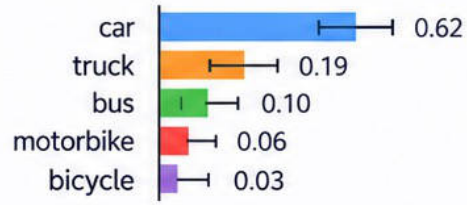


Aggregate predictions

Average the predictions over T stochastic forward passes:

$$\bar{p}(y \mid x) = \frac{1}{T} \sum_{t=1}^{T} p_t(y \mid x)$$

Mean prediction (with variation)



Uncertainty from disagreement

- Compute the variance or entropy of the predictions across passes.
- Higher disagreement → higher epistemic uncertainty.

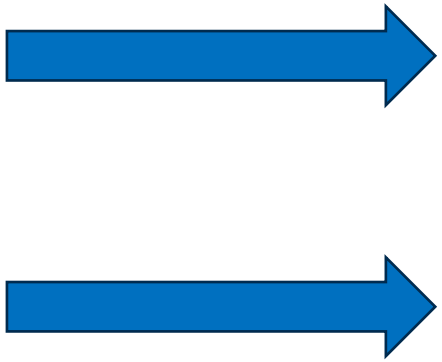
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Out-of-Distribution Detection

OOD Detection

- OOD detection is input-centric
- asks whether the input itself is atypical relative to training data
- can flag unfamiliar environments, corruptions, sensors, or novel classes
- examples:
  - max softmax baseline
  - feature distance
  - Energy score

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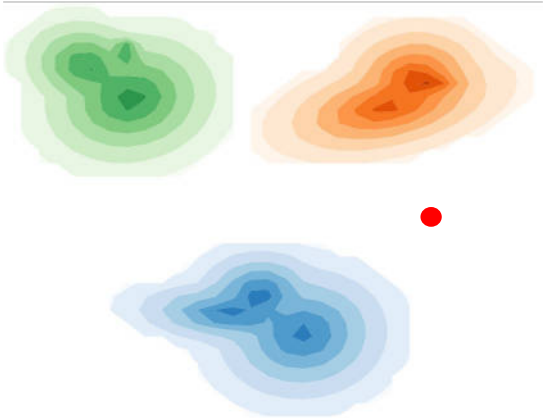
Common Ideas for OOD Detection

Maximum softmax probability

- very simple baseline
- low max probability → maybe unfamiliar
- but can still be confidently wrong

Feature-space distance

- compare an input's learned representation to training examples / class prototypes
- far away in feature space → potentially OOD



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Common Ideas for OOD Detection

- **Energy Score**
- operates directly on logits $z_k(x)$
- captures overall evidence for the known classes
- threshold the score to detect OOD

$$E(x) = -\log \sum_{k=1}^{K} \exp\big(z_k(x)\big)$$

- How is this different to Softmax?
  - Softmax looks at relative class probabilities and discards absolute logit scale
  - Two inputs can result in the same Softmax confidence but very different energy.

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OOD detection is not the same as failure detection

- An input can be OOD but still correctly classified
- An input can be in-distribution but still difficult / wrong
- High uncertainty, OOD, and prediction failure are related, but not identical

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Decision Making

We have estimated confidence, uncertainty and OOD-ness.

What should the robot actually do with that information?

Decision-Making

- A perception system can produce several signals:
  - prediction / class score
  - calibrated confidence
  - Uncertainty, OOD score
  - temporal / multi-view consistency

Decision-making has to combine estimated reliability with the consequences of being wrong.



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What can the robot do when it does not trust a prediction?

Uncertainty is only useful if it changes behaviour.

- **If reliability is high:** act normally
- **If reliability is uncertain:**
  - slow down
  - gather another observation
  - change viewpoint
  - use temporal evidence
  - consult another sensor / model
- **If reliability is low:**
  - abstain
  - enter a safe fallback behaviour
  - ask a human
  - stop / defer the action
- **If the problem persists:**
  - collect more data
  - adapt / retrain / fine-tune the model

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Selective prediction: Knowing when not to answer

A model does not have to make a prediction on every input!

if $r(x) < \tau$:
    predict
else:
    abstain

- $r(x)$ could be uncertainty, failure risk, an OOD score, etc.
- The threshold τ asks: *"How uncertain are we willing to be before the robot refuses to act?"*

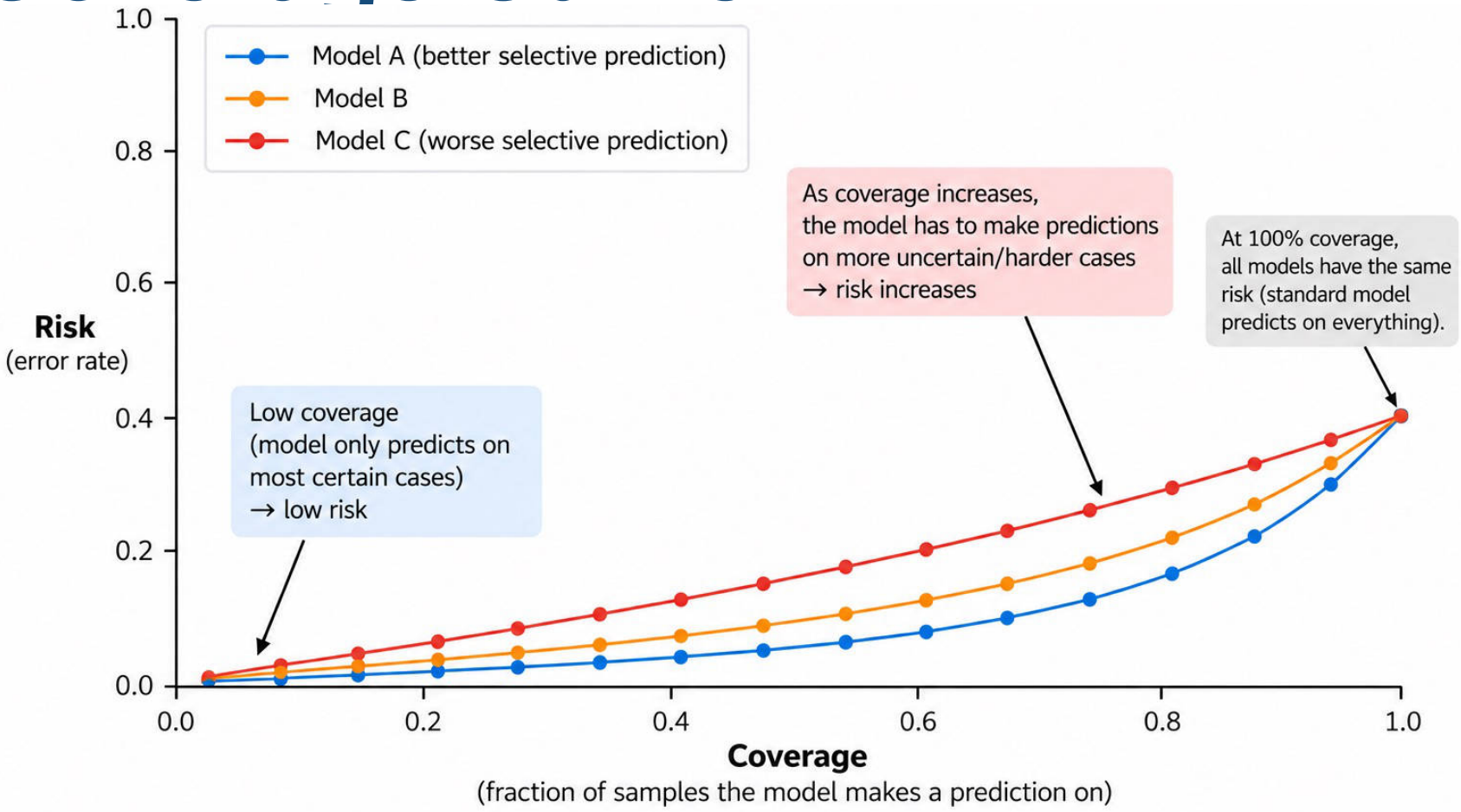
Two important concepts:

- **coverage** = fraction of inputs on which the model makes a prediction
- **risk** = error rate on the predictions it accepts

- Lower coverage can give lower risk.

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Risk-Coverage Curve



- **Coverage:** fraction of inputs for which the model makes a prediction (1.0 = predicts on everything).
- **Risk:** error rate on the accepted predictions.

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Right Response depends on Cost of Failure

Situation	Possible response
Warehouse object classification	Accept moderate uncertainty
Robot grasp	Acquire another view / replan
Obstacle detection	Favour high recall
Autonomous driving	Slow / stop / hand over

There is no universal uncertainty threshold.

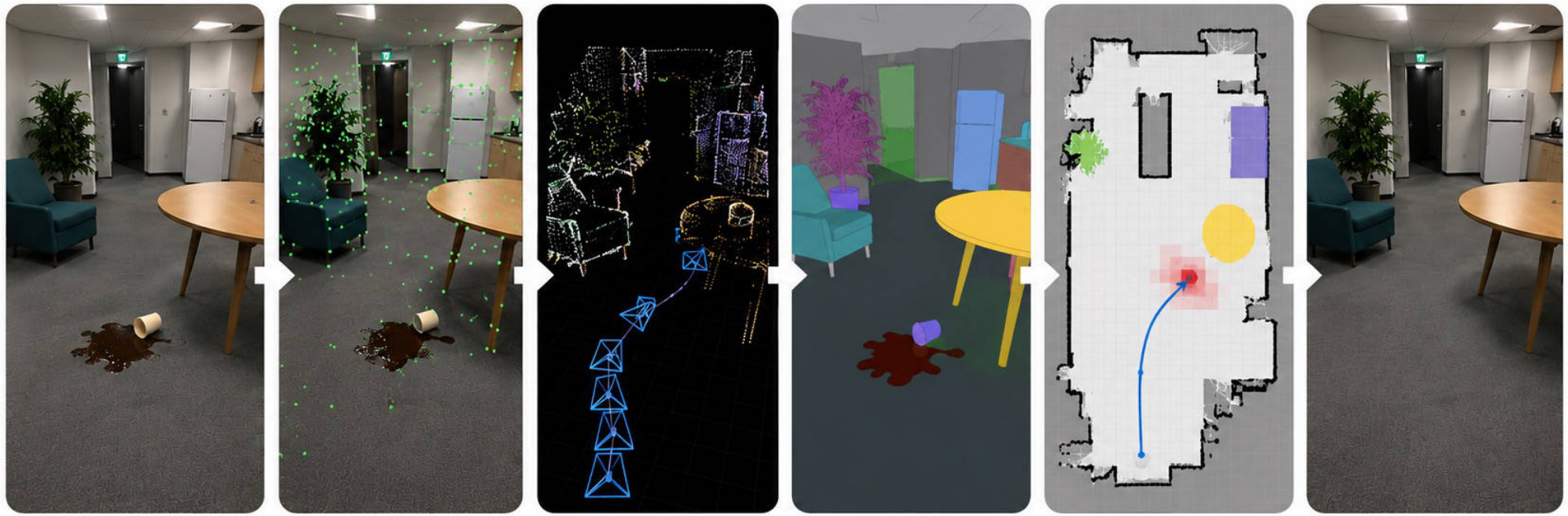
- The acceptable level of risk depends on:
  - consequence of failure
  - reversibility of the action
  - availability of additional information
  - time constraints
  - ability to recover

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Questions for Today?

Semester Recap

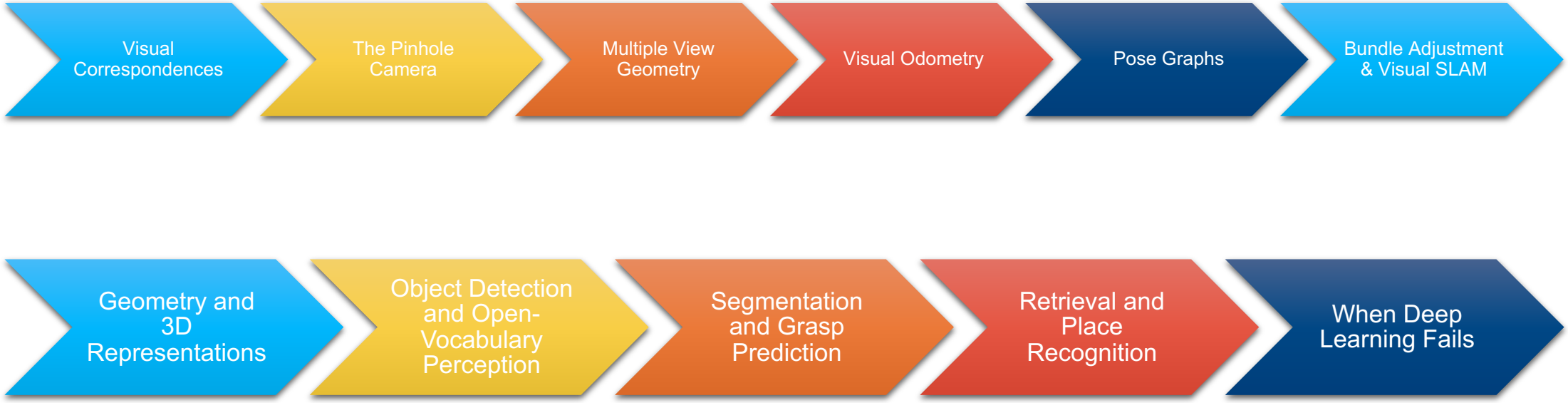
Robotic Vision is the use of cameras and algorithms to estimate information about the world that a robot needs for actions.



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Illustration by GPT5.5

Semester Recap



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Thanks for a Great Semester – Student Voice Survey –

I enjoyed teaching this semester, especially the interactions and questions during the lecture!

- If you enjoyed this unit, please let me know!
- If you noticed things we can improve, please also let me know!
- Your constructive feedback is important to us as educators.



STUDENT VOICE SURVEY

Your feedback is important, no matter how small.

Help to make positive changes to your learning experience.

Each unit survey is an entry into the weekly draw to win $250. You can win more than once, so get in quick!

A donation is made to QUT Guild Food Bank or Learning Potential Fund for every survey completed. So far we've raised $114,000.

It only takes a few minutes to complete.

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